SAAT / GAT QUANTITATIVE (ENGLISH) • UNIT 5.01

C SAAT Mathematical Logic, Matrices & Complex Numbers

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

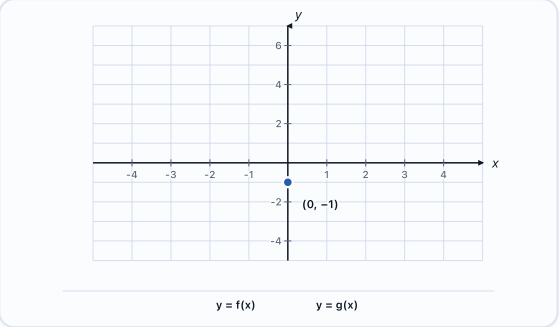
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 B	Q3 C	Q4 D	Q5 A
Q6 B	Q7 C	Q8 D	Q9 B	Q10 A
Q11 C	Q12 B	Q13 B	Q14 B	Q15 B
Q16 D	Q17 D	Q18 C	Q19 B	Q20 D
Q21 D	Q22 A	Q23 A	Q24 A	Q25 B

SAAT Mathematical Logic, Matrices & Complex Numbers Practice Exercises

1 MCQ

The coordinate diagram shows the parabolic curve $f_1(x) = x^2 - 14x + 64$ in the domain of SAAT Logic, Polynomials, Matrices & Complex Numbers. By completing the square $f_1(x) = (x - h)^2 + k$, find the sum $k + f_1(9)$.



A 34 ✓

B 41

C 19

D 32

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (SAAT Logic, Polynomials, Matrices & Complex Numbers):

1. Complete the square: $x^2 - 14x + 64 = (x - 7)^2 - 49 + 64 = (x - 7)^2 + 15$, giving minimum value $k = 15$. **[M1]**
2. Evaluate at $x = 9$: $f_1(9) = (2)^2 + 15 = 19$. **[M1]**
3. Sum the two quantities: $k + f_1(9) = 15 + 19 = 34$. **[A1]**

- **Final Verified Answer:** 34.

2 MCQ

A mathematical model in SAAT Logic, Polynomials, Matrices & Complex Numbers is defined by the cubic function $y_2(x) = 3x^3 - 7x^2 + 15x - 12$. Calculate the exact gradient of the tangent to the curve at the point where $x = 3$.

A 96

B 54 ✓

C 39

D 28

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (SAAT Logic, Polynomials, Matrices & Complex Numbers):

1. Differentiate term by term using the power rule: $\frac{dy}{dx} = 9x^2 - 14x + 15$. **[M1]**
2. Substitute $x = 3$: $\left.\frac{dy}{dx}\right|_{x=3} = 9(9) - 14(3) + 15 = 54$. **[M1][A1]**

- **Final Verified Answer:** 54.

3 MCQ

In SAAT Logic, Polynomials, Matrices & Complex Numbers, two displacement vectors are given by $\mathbf{u}_3 = \langle 7, -8 \rangle$ and $\mathbf{v}_3 = \langle 9, 7 \rangle$. Compute the exact scalar value of $\mathbf{u}_3 \cdot \mathbf{v}_3 + \|\mathbf{u}_3\|^2$.

A 7

B 113

C 120 ✓

D 232

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (SAAT Logic, Polynomials, Matrices & Complex Numbers):

1. Compute the inner product: $\mathbf{u}_3 \cdot \mathbf{v}_3 = (7)(9) + (-8)(7) = 63 + (-56) = 7$. **[M1]**
2. Compute the squared norm: $\|\mathbf{u}_3\|^2 = 7^2 + (-8)^2 = 113$. **[M1]**
3. Combine: $7 + 113 = 120$. **[A1]**

- **Final Verified Answer:** 120.

4 MCQ

The linear growth profile shown in the diagram for SAAT Logic, Polynomials, Matrices & Complex Numbers follows an arithmetic progression with first term $a_1 = 8$ and common difference $d = 6$. Find the exact sum S_{15} of the first 15 terms.

A 92

B 840

C 1380

D 750 ✓

WORKED SOLUTION & VERIFICATION

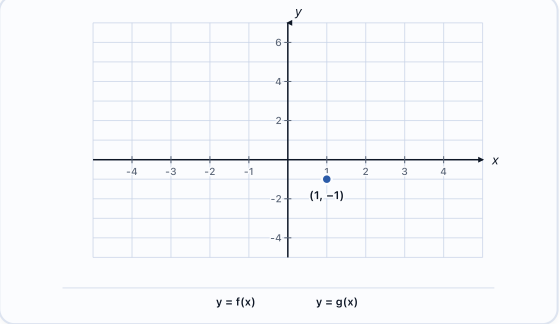
Mark Scheme & Examiner Solution (SAAT Logic, Polynomials, Matrices & Complex Numbers):

1. Use the sum formula for an arithmetic series: $S_n = \frac{n}{2}[2a_1 + (n-1)d]$. **[M1]**
2. Substitute $n = 15, a_1 = 8, d = 6$: $S_{15} = \frac{15}{2}[16 + 14 \times 6] = \frac{15}{2} \times 100 = 750$. **[M1][A1]**

- **Final Verified Answer:** 750.

5 MCQ

In SAAT Logic, Polynomials, Matrices & Complex Numbers, evaluate the exact definite integral $I_5 = \int_0^2 (18x^2 + 28x)\,dx$.



- **A** 104 ✓
- **B** 128
- **C** 132
- **D** 53

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (SAAT Logic, Polynomials, Matrices & Complex Numbers):

1. Integrate term by term: $\int (18x^2 + 28x)\,dx = [6x^3 + 14x^2]_0^2$. **[M1]**
2. Substitute the upper and lower limits: $(6(2)^3 + 14(2)^2) - 0 = 48 + 56 = 104$. **[M1][A1]**

- **Final Verified Answer:** 104.

6 MCQ

In SAAT Logic, Polynomials, Matrices & Complex Numbers, consider the quadratic function $f_6(x) = x^2 - 16x + 151$. Express $f_6(x)$ in the completed-square form $(x - h)^2 + k$ and compute the exact value of $k + f_6(15)$.

- **A** 231
- **B** 223 ✓
- **C** 136
- **D** 216

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (SAAT Logic, Polynomials, Matrices & Complex Numbers):

1. Complete the square: $x^2 - 16x + 151 = (x - 8)^2 - 64 + 151 = (x - 8)^2 + 87$, giving minimum value $k = 87$. **[M1]**
2. Evaluate at $x = 15$: $f_6(15) = (7)^2 + 87 = 136$. **[M1]**
3. Sum the two quantities: $k + f_6(15) = 87 + 136 = 223$. **[A1]**

- **Final Verified Answer:** 223.

7 MCQ

The graph illustrates the cubic model $y_7(x) = 3x^3 - 8x^2 + 33x - 19$ studied in SAAT Logic, Polynomials, Matrices & Complex Numbers. Find the instantaneous gradient $\frac{dy}{dx}$ of the tangent to the curve at $x = 8$.

A 609

B 448

C 481 ✓

D 89

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (SAAT Logic, Polynomials, Matrices & Complex Numbers):

1. Differentiate term by term using the power rule: $\frac{dy}{dx} = 9x^2 - 16x + 33$. **[M1]**
2. Substitute $x = 8$: $\left.\frac{dy}{dx}\right|_{x=8} = 9(64) - 16(8) + 33 = 481$. **[M1][A1]**

- **Final Verified Answer:** 481.

8 MCQ

In SAAT Logic, Polynomials, Matrices & Complex Numbers, two displacement vectors are given by $\mathbf{u}_8 = \langle 8, -7 \rangle$ and $\mathbf{v}_8 = \langle 8, 5 \rangle$. Compute the exact scalar value of $\mathbf{u}_8 \cdot \mathbf{v}_8 + \|\mathbf{u}_8\|^2$.

A 29

B 113

C 212

D 142 ✓

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (SAAT Logic, Polynomials, Matrices & Complex Numbers):

1. Compute the inner product: $\mathbf{u}_8 \cdot \mathbf{v}_8 = (8)(8) + (-7)(5) = 64 + (-35) = 29$. **[M1]**
2. Compute the squared norm: $\|\mathbf{u}_8\|^2 = 8^2 + (-7)^2 = 113$. **[M1]**
3. Combine: $29 + 113 = 142$. **[A1]**

- **Final Verified Answer:** 142.

9 MCQ

For what value of k does the matrix $A = \begin{pmatrix} 2 & 3 \\ 6 & k \end{pmatrix}$ fail to have an inverse (is singular)?

A 6

B 9 ✓

C -9

D 11

WORKED SOLUTION & VERIFICATION

A matrix is singular iff $\det(A) = 0$: $2k - (3)(6) = 0 \implies 2k = 18 \implies k = 9$.

10 MCQ

For what value of k does the matrix $A = \begin{pmatrix} 3 & 4 \\ 12 & k \end{pmatrix}$ fail to have an inverse (is singular)?

A 16 ✓

B -16

C 12

D 19

WORKED SOLUTION & VERIFICATION

A matrix is singular iff $\det(A) = 0$: $3k - (4)(12) = 0 \implies 3k = 48 \implies k = 16$.

11 MCQ

For what value of k does the matrix $A = \begin{pmatrix} 4 & 5 \\ 20 & k \end{pmatrix}$ fail to have an inverse (is singular)?

A -25	B 20
C 25 ✓	D 29

WORKED SOLUTION & VERIFICATION

A matrix is singular iff $\det(A) = 0$: $4k - (5)(20) = 0 \implies 4k = 100 \implies k = 25$.

12 MCQ

For what value of k does the matrix $A = \begin{pmatrix} 5 & 6 \\ 30 & k \end{pmatrix}$ fail to have an inverse (is singular)?

A 30	B 36 ✓
C 41	D -36

WORKED SOLUTION & VERIFICATION

A matrix is singular iff $\det(A) = 0$: $5k - (6)(30) = 0 \implies 5k = 180 \implies k = 36$.

13 MCQ

For what value of k does the matrix $A = \begin{pmatrix} 6 & 7 \\ 12 & k \end{pmatrix}$ fail to have an inverse (is singular)?

A 20	B 14 ✓
C 42	D -14

WORKED SOLUTION & VERIFICATION

A matrix is singular iff $\det(A) = 0$: $6k - (7)(12) = 0 \implies 6k = 84 \implies k = 14$.

14 MCQ

For what value of k does the matrix $A = \begin{pmatrix} 7 & 1 \\ 21 & k \end{pmatrix}$ fail to have an inverse (is singular)?

A -3	B 3 ✓
C 10	D 7

WORKED SOLUTION & VERIFICATION

A matrix is singular iff $\det(A) = 0$: $7k - (1)(21) = 0 \implies 7k = 21 \implies k = 3$.

15 MCQ

For what value of k does the matrix $A = \begin{pmatrix} 2 & 2 \\ 8 & k \end{pmatrix}$ fail to have an inverse (is singular)?

A 10	B 8 ✓
C -8	D 4

WORKED SOLUTION & VERIFICATION

A matrix is singular iff $\det(A) = 0$: $2k - (2)(8) = 0 \implies 2k = 16 \implies k = 8$.

16 MCQ

For what value of k does the matrix $A = \begin{pmatrix} 3 & 3 \\ 15 & k \end{pmatrix}$ fail to have an inverse (is singular)?

A 9	B -15
C 18	D 15 ✓

WORKED SOLUTION & VERIFICATION

A matrix is singular iff $\det(A) = 0$: $3k - (3)(15) = 0 \implies 3k = 45 \implies k = 15$.

17 MCQ

For what value of k does the matrix $A = \begin{pmatrix} 4 & 4 \\ 24 & k \end{pmatrix}$ fail to have an inverse (is singular)?

A 28	B 16
C -24	D 24 ✓

WORKED SOLUTION & VERIFICATION

A matrix is singular iff $\det(A) = 0$: $4k - (4)(24) = 0 \implies 4k = 96 \implies k = 24$.

18 MCQ

For what value of k does the matrix $A = \begin{pmatrix} 5 & 5 \\ 10 & k \end{pmatrix}$ fail to have an inverse (is singular)?

A 15	B -10
C 10 ✓	D 25

WORKED SOLUTION & VERIFICATION

A matrix is singular iff $\det(A) = 0$: $5k - (5)(10) = 0 \implies 5k = 50 \implies k = 10$.

19 MCQ

For what value of k does the matrix $A = \begin{pmatrix} 6 & 6 \\ 18 & k \end{pmatrix}$ fail to have an inverse (is singular)?

A 36	B 18 ✓
C 24	D -18

WORKED SOLUTION & VERIFICATION

A matrix is singular iff $\det(A) = 0$: $6k - (6)(18) = 0 \implies 6k = 108 \implies k = 18$.

20 MCQ

For what value of k does the matrix $A = \begin{pmatrix} 7 & 7 \\ 28 & k \end{pmatrix}$ fail to have an inverse (is singular)?

A 49	B 35
C -28	D 28 ✓

WORKED SOLUTION & VERIFICATION

A matrix is singular iff $\det(A) = 0$: $7k - (7)(28) = 0 \implies 7k = 196 \implies k = 28$.

21 MCQ

For what value of k does the matrix $A = \begin{pmatrix} 2 & 1 \\ 10 & k \end{pmatrix}$ fail to have an inverse (is singular)?

A 2	B -5
C 7	D 5 ✓

WORKED SOLUTION & VERIFICATION

A matrix is singular iff $\det(A) = 0$: $2k - (1)(10) = 0 \implies 2k = 10 \implies k = 5$.

22 MCQ

For what value of k does the matrix $A = \begin{pmatrix} 3 & 2 \\ 18 & k \end{pmatrix}$ fail to have an inverse (is singular)?

A 12 ✓	B -12
C 15	D 6

WORKED SOLUTION & VERIFICATION

A matrix is singular iff $\det(A) = 0$: $3k - (2)(18) = 0 \implies 3k = 36 \implies k = 12$.

23 MCQ

For what value of k does the matrix $A = \begin{pmatrix} 4 & 3 \\ 8 & k \end{pmatrix}$ fail to have an inverse (is singular)?

- **A** 6 ✓
- **B** -6
- **C** 12
- **D** 10

WORKED SOLUTION & VERIFICATION

A matrix is singular iff $\det(A) = 0$: $4k - (3)(8) = 0 \implies 4k = 24 \implies k = 6$.

24 MCQ

For what value of k does the matrix $A = \begin{pmatrix} 5 & 4 \\ 15 & k \end{pmatrix}$ fail to have an inverse (is singular)?

- **A** 12 ✓
- **B** 20
- **C** 17
- **D** -12

WORKED SOLUTION & VERIFICATION

A matrix is singular iff $\det(A) = 0$: $5k - (4)(15) = 0 \implies 5k = 60 \implies k = 12$.

25 MCQ

For what value of k does the matrix $A = \begin{pmatrix} 6 & 5 \\ 24 & k \end{pmatrix}$ fail to have an inverse (is singular)?

- **A** -20
- **B** 20 ✓
- **C** 30
- **D** 26

WORKED SOLUTION & VERIFICATION

A matrix is singular iff $\det(A) = 0$: $6k - (5)(24) = 0 \implies 6k = 120 \implies k = 20$.